Shipment route selection under cost, time, and risk

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Repository: https://github.com/vivekvar-dl/aiqon-route-optimizer

1. Decision framework

Routeplan selects among supplied candidate routes. Optimal means the lowest risk-adjusted landed cost among routes whose quoted schedule meets the shipment deadline. Inputs include origin, destination, cargo value, priority, deadline, and leg-level freight, time, delay probability, congestion, and geopolitical risk. A CLI and Streamlit interface share the same Python engine.

For scenario s, F is freight, V cargo value, L(s) cargo loss, T(s) transit hours, and D the deadline. Holding price h and lateness price p are fractions of cargo value per day:

```
C(s) = F + L(s) + h * V * T(s) / 24
                + p * V * max(T(s) - D, 0) / 24
Score = mean(C) + a * [CVaR90(C) - mean(C)]
```

CVaR90 is the mean of the worst 10% of simulated costs. Tail weight a blends expected cost toward this tail mean. Lowest score wins; ties break on freight plus expected loss, then route letter. All score components are dollars, so explanations reconcile extra freight against holding, lateness, and tail savings. Unlike alternative-dependent normalization, these prices retain their economic meaning when a candidate is added.

Priority	Holding / day	Lateness / day	Tail weight
Cost-first	0.03% of V	0.2% of V	0.10
Balanced	0.10% of V	1% of V	0.30
Time-critical	0.30% of V	5% of V	0.30
Risk-averse	0.10% of V	2% of V	0.90

Prices are explicit judgments, not learned coefficients or contractual quotes. Eligibility uses scheduled hours; stochastic lateness is priced. If every schedule misses the deadline, the engine abstains and reports the fastest schedule's shortfall. There is no minimum on-time probability constraint or delivery guarantee.

2. Synthetic data and outcome model

The seed-7 catalog contains 30 shipments and 120 two-leg routes, cycling through 3-5 options. Twelve origin/destination cities and six maritime gateways form a coarse sea-corridor graph. Sea distances follow manually specified waypoints with a 5% allowance; air distances use great-circle geometry. Candidates include sea, air, sea-air, and air-sea alternatives, with transfer hubs chosen by distance rules. No ocean-crossing rail is generated. Port-to-airport handling is abstracted into dwell time. These are illustrative corridors, not verified carrier services or navigational charts; construction and source context are in docs/dataset.md.

Sea costs $350 + $0.045/km per leg at 32 km/h plus 16 hours dwell; air costs $900 + $0.55/km at 780 km/h plus 5 hours dwell. Cargo values range from $80,000 to $1,500,000. Every tenth shipment deliberately has an impossible deadline. Risk scores and prices are synthetic per-consignment assumptions, with fixed cargo size implicit.

Each leg draws a delay at its supplied probability, with exponential delay length averaging leg hours * (0.15 + 0.60 * congestion). A disruption occurs with probability 0.08 * geopolitical risk, adds 96 hours, and damages 30% of remaining cargo with conditional probability 0.25. Risk labels affect time and loss rather than being charged again. Shared hops reuse draws within a shipment; nominal unrelated legs are independent. Seeded planning and held-out evaluation use separate streams.

3. Explanations and reliability

The interface names hubs and modes; the explanation contrasts quotes and schedules and reconciles the score in dollars. It reports deadline slack, simulated on-time probability, a 95% Wilson sampling interval, and 90th-percentile arrival time. An interval covers sampling error under the assumed simulator; it does not cover model error. Pareto labels compare freight plus expected loss, expected hours, and late probability. They are diagnostics, not vetoes: those summaries do not capture tail severity.

For SHP-005, Dubai to Los Angeles, cargo is $86,947 and deadline 669 hours. Cost-first selects D, air then sea via Shanghai. D has 233.81 hours of scheduled slack, yet only 82.8% simulated on-time arrivals (interval 81.1%-84.4%); its 90th-percentile arrival is 798.3 hours. Balanced instead chooses air route B via Hamburg. B pays $3,993 more in freight plus expected loss but saves $1,884 in holding, $1,548 in lateness, and $4,476 in tail risk: a rounded $3,915 score advantage. The example separates schedule eligibility from delivery reliability and shows how preference changes alter a valid choice.

Price sensitivity independently scales each economic price by 0.5, 0.67, 0.8, 1.25, 1.5, and 2, reusing outcomes; duplicate capped tail weights are skipped. Profile overrides expose different business priorities. These probes measure preference sensitivity, not Monte Carlo ranking uncertainty.

4. Evaluation with matching constraints

Planning uses 2,000 scenarios per route. Nominal evaluation uses 1,000 held-out scenarios per route; competing policies share outcomes. Cheapest-feasible and fastest-feasible obey the same scheduled deadline. Hindsight references separately minimize held-out mean cost and the mean-CVaR score among eligible routes. Mean-cost regret and risk-score regret are averaged per shipment against their corresponding references. Oracles see held-out samples and are diagnostic bounds, not deployable policies.

Policy, 27 feasible shipments	Total mean cost	On-time	Mean regret	Risk regret
Recommended	$213,799	98.30%	$14.80	$52.11
Cheapest-feasible	$1,067,449	76.00%	$31,631.48	$80,816.17
Fastest-feasible	$223,586	100.00%	$377.31	$329.52
Mean-cost oracle-feasible	$213,399	97.11%	$0.00	$80.89

The mean-cost oracle has positive risk-score regret because it optimizes a different objective. Likewise, paying a mean-cost premium can be justified by tail savings. Sampled 100% arrivals do not establish certainty. Results demonstrate the authored decision model, not observed commercial savings.

Coverage is 27/30 (90%). The full-catalog scorecard flags a fastest-schedule fallback for infeasible shipments; this is evaluation-only and excluded above. Unconstrained baselines remain separate. Summed risk scores are shipment-score totals, not portfolio CVaR; cost_ratio is a mean of per-shipment ratios. Across five catalogs (seeds 7, 11, 21, 42, 99), recommended on-time rates are 95.35%-98.30%, with mean risk-score regret $14.68-$118.45. Tests cover closed-form choices, abstention, shared draws, fragile rankings, stress monotonicity, and interface behavior. CI verifies seeded result reproduction on Python 3.12 and 3.13.

5. Sampling stability and model-error stress

A second experiment keeps the seed-7 catalog fixed. Twenty independent planning seeds (8-27), each with 2,000 scenarios, reproduce the original seed-7 pick 86.85% of the time on average across 27 feasible shipments. Seven fall below a heuristic 90% agreement threshold: SHP-003, 006, 009, 012, 015, 018, and 027. SHP-018 selects its original D in only 6/20 runs; B wins 14/20. SHP-005 retains D in all runs. The interface exposes counts and flags low agreement for review. It does not claim statistical certification or silently replace a choice.

Seed-7 price-probe stability is 96.09%, higher than sampling agreement. The measures answer different questions. Rare cargo-loss draws can change tail scores, particularly for nearly tied routes; a narrow on-time interval alone cannot establish ranking stability. More scenarios or operator review are warranted before relying on such a choice.

Stress freezes the original recommendations and uses 5,000 paired held-out scenarios per route. Longer delays multiply delay length by 1.5. Underestimated events multiply delay probability by 1.5 and disruption probability by 2. Regional stress adds a 72-hour hold with 10% probability per touched region, shared across routes and shipments and applied once per unique region. Different regions remain independent. Combined stress applies all changes; event probabilities cap at 1. The nominal planner never sees these changed environments.

Environment	Total mean cost	On-time	Mean risk regret
Nominal	$220,052	98.27%	$145.75
Longer delays	$223,285	97.25%	$272.91
Underestimated events	$241,737	97.36%	$151.09
Regional holds	$236,398	97.80%	$217.53
Combined	$263,471	95.17%	$425.13

Combined stress raises mean cost 19.73% and reduces on-time arrivals by 3.10 percentage points. These are diagnostic stress assumptions, not calibrated forecasts. Compare within this experiment: its nominal estimates differ from the 1,000-scenario evaluation because rare-event sampling differs. Stress Pareto labels retain nominal planning criteria. Full results are in results/robustness.json and results/evaluation.json.

6. Limits and practical extension

Synthetic outcomes cannot validate real-world accuracy. Prices and probabilities are assumed; capacity, timetables, closures, customs, cargo dimensions, and carrier-specific booking quotes are omitted. Identical-hop fields assume one booking context. Regional holds introduce correlation but no event timeline, and per-shipment optimization does not optimize a joint portfolio tail objective. Wilson intervals do not cover uncertainty in estimated model parameters or scores. Twenty sampling runs and hand-selected stresses expose weaknesses without proving robustness.

The assignment permits optimization without ML. A useful extension would learn booking-time delay probabilities or arrival distributions from compatible historical legs, validate on later dates and unseen lanes, check leakage and calibration, and feed predictions into the existing optimizer. A model trained on synthetic outcomes only learns the authored simulator. Strict service-level contracts would also require chance constraints and an explicit escalation policy.

7. Reproduction and deliverables

README.md provides installation and CLI/demo instructions. Run pytest, scripts/run_evaluation.py, and scripts/run_robustness.py after installing the test extras. Generate the PDF with scripts/render_report.py using the report extra. The repository includes catalog, code, tests, both result artifacts, and this three-page report. docs/demo-script.md gives an optional recording outline under five minutes; no video is included.